Format Adherence Analysis

How closely do model responses follow the expected chain-of-thought format (paper §VI-C), and does adherence correlate with prediction correctness?

Expected Sections

1. **Card Description**
2. **Order of Events**
3. **Synergy Analysis**
4. **Conclusion**
5. **Final Score**

> **Adherence score**: fraction of expected sections found (0–1, continuous).
> **Order score**: are found sections in the right relative order? (0–1).
> **Correlation**: point-biserial r between adherence and binary correctness.

---

gpt4o

	Value
Pairs analyzed	5,625
Accuracy	80.5% (4,529 correct / 1,096 wrong)
Mean adherence (all)	1.000 ± 0.000
Variance in adherence (all)	0.0000

Mean adherence — correct	1.000 (var=0.0000)
Mean adherence — incorrect	1.000 (var=0.0000)
Adherence delta (correct − incorrect)	+0.000

Correlation with Correctness

Metric	r	p-value	Significance
Adherence score	nan	nan	N/A
Combined score	nan	nan	N/A
Response length	-0.1786	1.543e-41	***

Section Presence Rates

Section	Correct	Incorrect	Δ
Card Description	100.0%	100.0%	+0.0% —
Order of Events	100.0%	100.0%	+0.0% —
Synergy Analysis	100.0%	100.0%	+0.0% —
Conclusion	100.0%	100.0%	+0.0% —
Final Score	100.0%	100.0%	+0.0% —

---

gpt54

	Value

Pairs analyzed	5,625
Accuracy	80.1% (4,507 correct / 1,118 wrong)
Mean adherence (all)	0.313 ± 0.124
Variance in adherence (all)	0.0154
Mean adherence — correct	0.306 (var=0.0147)
Mean adherence — incorrect	0.341 (var=0.0169)
Adherence delta (correct – incorrect)	-0.035

Correlation with Correctness

Metric	r	p-value	Significance
Adherence score	-0.1134	1.437e-17	***
Combined score	-0.1211	7.678e-20	***
Response length	-0.3101	1.266e-125	***

Section Presence Rates

Section	Correct	Incorrect	Δ
Card Description	38.1%	46.0%	-7.9% ↓
Order of Events	6.7%	17.3%	-10.5% ↓
Synergy Analysis	6.7%	5.7%	+1.0% —

Conclusion	1.3%	1.5%	-0.2% —
Final Score	100.0%	100.0%	+0.0% —

---

gemini10pro

	Value
Pairs analyzed	5,625
Accuracy	42.9% (2,415 correct / 3,210 wrong)
Mean adherence (all)	0.992 ± 0.047
Variance in adherence (all)	0.0022
Mean adherence — correct	0.992 (var=0.0018)
Mean adherence — incorrect	0.992 (var=0.0025)
Adherence delta (correct – incorrect)	+0.001

Correlation with Correctness

Metric	r	p-value	Significance
Adherence score	0.0091	0.4971	ns
Combined score	0.0078	0.5587	ns

Response length	-0.2236	1.079e-64	***

Section Presence Rates

Section	Correct	Incorrect	Δ
Card Description	99.1%	99.2%	-0.1% —
Order of Events	99.1%	99.3%	-0.2% —
Synergy Analysis	98.1%	98.0%	+0.1% —
Conclusion	99.9%	99.5%	+0.4% —
Final Score	99.9%	99.7%	+0.2% —

---

gemini15flash

	Value
Pairs analyzed	4,810
Accuracy	75.5% (3,632 correct / 1,178 wrong)
Mean adherence (all)	1.000 ± 0.012
Variance in adherence (all)	0.0001
Mean adherence — correct	1.000 (var=0.0002)
Mean adherence — incorrect	1.000 (var=0.0000)

Adherence delta (correct – incorrect)	-0.000

Correlation with Correctness

Metric	r	p-value	Significance
Adherence score	-0.0082	0.5691	ns
Combined score	-0.0081	0.573	ns
Response length	-0.3031	9.87e-103	***

Section Presence Rates

Section	Correct	Incorrect	Δ
Card Description	100.0%	100.0%	-0.0% —
Order of Events	100.0%	100.0%	-0.0% —
Synergy Analysis	100.0%	100.0%	-0.0% —
Conclusion	100.0%	100.0%	-0.0% —
Final Score	100.0%	100.0%	+0.0% —

---

gpt4omini

	Value
Pairs analyzed	5,625
Accuracy	68.1% (3,832 correct / 1,793 wrong)
Mean adherence (all)	1.000 ± 0.008

Variance in adherence (all)	0.0001
Mean adherence — correct	1.000 (var=0.0000)
Mean adherence — incorrect	1.000 (var=0.0002)
Adherence delta (correct − incorrect)	+0.000

Correlation with Correctness

Metric	r	p-value	Significance
Adherence score	0.0195	0.1438	ns
Combined score	0.0195	0.1438	ns
Response length	-0.1883	4.448e-46	***

Section Presence Rates

Section	Correct	Incorrect	Δ
Card Description	100.0%	100.0%	+0.0% —
Order of Events	100.0%	100.0%	+0.0% —
Synergy Analysis	100.0%	99.9%	+0.1% —
Conclusion	100.0%	99.9%	+0.1% —
Final Score	100.0%	99.9%	+0.1% —

---

gpt4ominift

	Value
Pairs analyzed	5,625
Accuracy	79.4% (4,465 correct / 1,160 wrong)
Mean adherence (all)	1.000 ± 0.004
Variance in adherence (all)	0.0000
Mean adherence — correct	1.000 (var=0.0000)
Mean adherence — incorrect	1.000 (var=0.0000)
Adherence delta (correct – incorrect)	-0.000

Correlation with Correctness

Metric	r	p-value	Significance
Adherence score	-0.0096	0.471	ns
Combined score	-0.0105	0.4301	ns
Response length	-0.1931	2.115e-48	***

Section Presence Rates

Section	Correct	Incorrect	Δ
Card Description	100.0%	100.0%	+0.0% —
Order of Events	100.0%	100.0%	+0.0% —

Synergy Analysis	100.0%	100.0%	-0.0% —
Conclusion	100.0%	100.0%	+0.0% —
Final Score	100.0%	100.0%	+0.0% —

gemini3flash

	Value
Pairs analyzed	5,625
Accuracy	82.3% (4,628 correct / 997 wrong)
Mean adherence (all)	0.998 ± 0.027
Variance in adherence (all)	0.0007
Mean adherence — correct	0.999 (var=0.0004)
Mean adherence — incorrect	0.994 (var=0.0022)
Adherence delta (correct – incorrect)	+0.005

Correlation with Correctness

Metric	r	p-value	Significance
Adherence score	0.0690	2.213e-07	***

Combined score	0.0800	1.902e-09	***
Response length	-0.2367	1.719e-72	***

Section Presence Rates

Section	Correct	Incorrect	Δ
Card Description	99.9%	99.1%	+0.8% —
Order of Events	99.7%	99.1%	+0.6% —
Synergy Analysis	99.9%	99.6%	+0.3% —
Conclusion	99.9%	99.1%	+0.8% —
Final Score	100.0%	100.0%	+0.0% —

Response Length Analysis

Across all models, incorrect predictions have significantly longer responses.
This is a confound: models write more when pairs are harder (true synergies/anti-synergies),
and harder pairs remain hard to classify regardless of response length.

Cross-Model Summary

Model	Acc	Mean (correct)	Mean (incorrect)	Diff	r	d	p (MWU)	sig
gpt4o	80.5%	1329 (σ=174)	1409 (σ=177)	-80	-0.1786	-0.458	1.15e-40	***
gpt54	80.1%	804 (σ=222)	992 (σ=266)	-189	-0.3101	-0.817	1.7e-106	***
gemini10pro	42.9%	1346 (σ=293)	1486 (σ=309)	-140	-0.2236	-0.463	1.69e-72	***
gemini15flash	75.5%	1127 (σ=215)	1290 (σ=236)	-163	-0.3031	-0.739	1.41e-97	***
gpt4omini	68.1%	1375 (σ=184)	1452 (σ=189)	-76	-0.1883	-0.411	1.19e-45	***
gpt4ominift	79.4%	714 (σ=107)	768 (σ=124)	-54	-0.1931	-0.486	1.16e-42	***
gemini3flash	82.3%	1651 (σ=286)	1836 (σ=310)	-185	-0.2367	-0.638	1.71e-69	***

gpt4o — detail

Length by outcome

	Mean	Median	Std
Correct	1329	1318	174
Incorrect	1409	1397	177

- r(length, correct) = **-0.1786**, p=1.54e-41 ***
- Cohen's d = **-0.458** (correct vs incorrect)
- r^2 = 0.0319 (3.19% variance explained)

Accuracy by length quartile (Q1=shortest, Q4=longest)

Quartile	n	Mean length	Accuracy
Q1	1412	1132	89.8%
Q2	1405	1277	82.7%
Q3	1404	1391	79.1%
Q4	1404	1579	70.4%

Confound check: mean length and accuracy by GT class

GT class	n	Mean length	Accuracy
Anti-syn (-1)	125	1387	31.2%
Neutral (0)	4553	1339	83.8%
Synergy (+1)	947	1363	71.1%

gpt54 — detail

Length by outcome

	Mean	Median	Std
Correct	804	767	222
Incorrect	992	958	266

- r(length, correct) = **-0.3101**, p=1.27e-125 ***

- Cohen's d = **-0.817** (correct vs incorrect)
- r^2 = 0.0961 (9.61% variance explained)

Accuracy by length quartile (Q1=shortest, Q4=longest)

Quartile	n	Mean length	Accuracy
Q1	1409	573	92.8%
Q2	1409	731	86.3%
Q3	1402	886	78.4%
Q4	1405	1176	62.9%

Confound check: mean length and accuracy by GT class

GT class	n	Mean length	Accuracy
Anti-syn (-1)	125	985	34.4%
Neutral (0)	4553	825	84.1%
Synergy (+1)	947	900	66.9%

gemini10pro — detail

Length by outcome

	Mean	Median	Std
Correct	1346	1307	293
Incorrect	1486	1444	309

- r(length, correct) = **-0.2236**, p=1.08e-64 ***
- Cohen's d = **-0.463** (correct vs incorrect)
- r^2 = 0.0500 (5.00% variance explained)

Accuracy by length quartile (Q1=shortest, Q4=longest)

Quartile	n	Mean length	Accuracy
Q1	1413	1080	61.1%

Q2	1403	1300	43.4%
Q3	1405	1486	38.0%
Q4	1404	1838	29.1%

Confound check: mean length and accuracy by GT class

GT class	n	Mean length	Accuracy
Anti-syn (-1)	125	1507	27.2%
Neutral (0)	4553	1422	35.6%
Synergy (+1)	947	1432	80.1%

gemini15flash — detail

Length by outcome

	Mean	Median	Std
Correct	1127	1103	215
Incorrect	1290	1280	236

- r(length, correct) = **-0.3031**, p=9.87e-103 ***
- Cohen's d = **-0.739** (correct vs incorrect)
- r^2 = 0.0918 (9.18% variance explained)

Accuracy by length quartile (Q1=shortest, Q4=longest)

Quartile	n	Mean length	Accuracy
Q1	1209	905	89.7%
Q2	1197	1071	82.2%
Q3	1202	1213	76.0%
Q4	1202	1480	54.2%

Confound check: mean length and accuracy by GT class

GT class	n	Mean length	Accuracy

Anti-syn (-1)	109	1284	29.4%
Neutral (0)	3892	1150	80.4%
Synergy (+1)	809	1232	58.0%

gpt4omini — detail

Length by outcome

	Mean	Median	Std
Correct	1375	1368	184
Incorrect	1452	1448	189

- r(length, correct) = **-0.1883**, p=4.45e-46 ***
- Cohen's d = **-0.411** (correct vs incorrect)
- r^2 = 0.0355 (3.55% variance explained)

Accuracy by length quartile (Q1=shortest, Q4=longest)

Quartile	n	Mean length	Accuracy
Q1	1412	1166	78.8%
Q2	1401	1332	71.6%
Q3	1418	1457	66.1%
Q4	1394	1646	55.9%

Confound check: mean length and accuracy by GT class

GT class	n	Mean length	Accuracy
Anti-syn (-1)	125	1491	22.4%
Neutral (0)	4553	1401	69.1%
Synergy (+1)	947	1378	69.3%

gpt4ominift — detail

Length by outcome

	Mean	Median	Std
Correct	714	703	107
Incorrect	768	757	124

- r(length, correct) = **-0.1931**, p=2.11e-48 ***
- Cohen's d = **-0.486** (correct vs incorrect)
- r^2 = 0.0373 (3.73% variance explained)

Accuracy by length quartile (Q1=shortest, Q4=longest)

Quartile	n	Mean length	Accuracy
Q1	1411	593	87.2%
Q2	1409	680	83.8%
Q3	1399	750	78.0%
Q4	1406	876	68.4%

Confound check: mean length and accuracy by GT class

GT class	n	Mean length	Accuracy
Anti-syn (-1)	125	770	16.0%
Neutral (0)	4553	720	86.8%
Synergy (+1)	947	743	52.2%

gemini3flash — detail

Length by outcome

	Mean	Median	Std
Correct	1651	1602	286
Incorrect	1836	1783	310

- r(length, correct) = **-0.2367**, p=1.72e-72 ***
- Cohen's d = **-0.638** (correct vs incorrect)
- r^2 = 0.0560 (5.60% variance explained)

Accuracy by length quartile (Q1=shortest, Q4=longest)

Quartile	n	Mean length	Accuracy
Q1	1408	1353	93.5%
Q2	1406	1553	86.3%
Q3	1405	1733	80.1%
Q4	1406	2097	69.3%

Confound check: mean length and accuracy by GT class

GT class	n	Mean length	Accuracy
Anti-syn (-1)	125	1832	48.0%
Neutral (0)	4553	1677	84.5%
Synergy (+1)	947	1698	76.0%

Length analysis generated by `length_analysis.py`

Decimal Ground Truth Accuracy Analysis

The decimal GT rates pairs on a finer scale (–2 to +2 in 0.5 steps). Models only predict 3-class labels (–1, 0, +1), so this measures how well they handle pairs whose true synergy strength is borderline (GT=±0.5) vs definitive (GT=±1.0/1.5/2.0).

Value Distribution (75×75 = 5,625 pairs)

Value	Count	%
–2.0	1	0.0%
–1.5	1	0.0%
–1.0	17	0.3%
–0.5	110	2.0%
0.0	4,588	81.5%

+0.5	354	6.3%
+1.0	419	7.4%
+1.5	46	0.8%
+2.0	89	1.6%

Mask Definitions

Mask	Decimal GT values	Expected prediction	n
`neutral`	[0.0]	0	4,588
`positive_half`	[0.5]	+1	354
`positive_clear`	[1.0, 1.5, 2.0]	+1	554
`negative_half`	[–0.5]	–1	110
`negative_clear`	[–1.0, –1.5, –2.0]	–1	19

> 1.5 is treated as equally definitive as 1.0 and 2.0 — only 0.5 is the true borderline.
> Accuracy and F1 are computed against the **real integer GT** (–1/0/+1); the decimal GT is used only to select which pairs enter each mask.

Borderline Positive Pairs (GT=0.5, n=354)

Model	Accuracy	F1 (target)	Macro F1	vs. clear positives
gemini10pro	71.5%	79.5%	14.5%	88.3% (–16.8pp)
gemini3flash	63.6%	77.7%	33.5%	88.6% (–25.0pp)
gpt4o	54.0%	70.8%	29.7%	85.7% (–31.7pp)
gpt4omini	49.7%	65.3%	31.6%	84.1% (–34.4pp)
gpt54	46.0%	63.0%	27.0%	85.2% (–39.2pp)
gpt4ominift	33.9%	53.3%	22.9%	65.2% (–31.3pp)
gemini15flash	28.5%	44.8%	22.3%	65.5% (–37.0pp)

Borderline Negative Pairs (GT=–0.5, n=110)

Model	Accuracy	F1 (target)	Macro F1	vs. clear negatives
gemini3flash	51.8%	60.9%	39.2%	84.2% (–32.4pp)

gpt54	39.1%	43.9%	30.6%	63.2% (–24.1pp)
gpt4o	26.4%	38.0%	27.9%	73.7% (–47.3pp)
gemini10pro	25.5%	38.4%	25.6%	42.1% (–16.6pp)
gemini15flash	23.6%	33.0%	27.4%	47.4% (–23.8pp)
gpt4omini	21.8%	25.3%	19.4%	52.6% (–30.8pp)
gpt4ominift	15.5%	20.5%	21.8%	47.4% (–31.9pp)

Clear Positive Pairs (GT=1.0/1.5/2.0, n=554)

Model	Accuracy	F1 (target)	Macro F1
gemini3flash	88.6%	94.1%	32.6%
gemini10pro	88.3%	93.9%	23.5%
gpt4o	85.7%	92.4%	31.7%
gpt54	85.2%	92.1%	31.6%
gpt4omini	84.1%	91.3%	30.4%
gpt4ominift	65.2%	79.0%	26.7%
gemini15flash	65.5%	79.3%	26.8%

Clear Negative Pairs (GT=–1.0/–1.5/–2.0, n=19)

Model	Accuracy	F1 (target)	Macro F1
gemini3flash	84.2%	88.2%	29.4%
gpt4o	73.7%	87.5%	29.2%
gpt54	63.2%	73.3%	24.4%
gpt4omini	52.6%	71.4%	23.8%
gemini15flash	47.4%	66.7%	38.9%
gpt4ominift	47.4%	66.7%	44.4%
gemini10pro	42.1%	61.5%	27.2%

Half vs Clear Summary

Model	Pos half (0.5)	Pos clear (1.0–2.0)	Neg half (–0.5)	Neg clear (–1.0––2.0)

gemini3flash	63.6%	88.6%	**51.8%**	**84.2%**
gemini10pro	**71.5%**	88.3%	25.5%	42.1%
gpt4o	54.0%	85.7%	26.4%	73.7%
gpt54	46.0%	85.2%	39.1%	63.2%
gpt4omini	49.7%	84.1%	21.8%	52.6%
gpt4ominift	33.9%	65.2%	15.5%	47.4%
gemini15flash	28.5%	65.5%	23.6%	47.4%

Key findings:

- All models drop 15–40pp on borderline positives vs clear positives — 0.5-rated pairs are a genuine weak spot across the board
- Negative borderlines (–0.5) are catastrophically hard; only gemini3flash exceeds coin-flip (51.8%), gpt4ominift bottoms out at 15.5%
- gemini3flash leads on every negative mask — the only model that meaningfully detects weak anti-synergies
- gemini10pro's high borderline-positive accuracy (71.5%) is striking given its poor overall accuracy (42.9%) — it over-predicts +1 broadly, which happens to pay off here
- gpt4ominift and gemini15flash share the same clear-positive accuracy (65.2%/65.5%) despite very different architectures — both struggle with weaker synergies
- The fine-tuned mini model (gpt4ominift) is worst on all negative masks despite being the strongest on neutrals (86.7%), consistent with over-fitting to the dominant class

---

Decimal analysis generated by `decimal_accuracy.py`